PART V

Alternative Modes

Gentler paths when full autoISF FCL isn't the right fit — hybrid with Meal Announcement, simpler FCL via AAPS Automations.

CHAPTER 7

Advanced HCL with Meal Announcement

Full Closed Loop is not the only good destination. For many users — and especially for children, or for adults who lack one of the FCL prerequisites — a smarter hybrid loop with Meal Announcement (MA) is a better fit.

This chapter describes how to use autoISF in a hybrid loop where you still give a bolus at meal time, but skip carb counting. Most of the setup overlaps with FCL. The differences are what this chapter is about.

7.1 What Meal Announcement is, and who it's for

Meal Announcement (MA) mode: you give a small pre-meal bolus (or a full meal bolus), but you do not enter carbs. autoISF's ISF modulation does the rest.

It sits in the middle ground:

- More than HCL: less per-meal thinking than full HCL, because you don't need to count carbs or calculate bolus sizes precisely.
- Less than FCL: you still bolus at meal time, so you don't get the hands-off experience.

WHO IT'S A GOOD FIT FOR

- Children, where Lyumjev tolerance, basal rates, or caregiver coordination make full FCL risky.
- Anyone who can't tolerate Lyumjev or Fiasp well — MA accommodates slower insulins because you're bolusing.
- Users without perfect Bluetooth discipline — MA gives you a safety margin when a few minutes of lost connection won't immediately sink the meal.
- Users with leaking pods or occasional pump reliability issues — MA tolerates these better than FCL.
- Users with jumpy CGMs — MA doesn't need aggressive early acceleration detection.
- Users with very low hourly basal — MA doesn't require super-boosted SMBs.
- Anyone transitioning who isn't ready to commit to full FCL.

MA can also be a staging post. Set up your system for MA first; move to full FCL later if and when the missing prerequisites fall into place.

7.2 Hurdles for FCL, and how MA bridges them

FCL deficit	*Bridging via MA*
Lyumjev/Fiasp not tolerated, or frequent occlusions	Pre-bolus with a different insulin, even via pen (with manual AAPS entry). Different cannulas / pods may help.
Poor 24/7 Bluetooth discipline	The meal bolus + profile basal cover most of the meal even if Bluetooth drops.
Leaking pods	Pre-bolus (via pen if needed) bypasses the worst failure moments.
Jumpy CGM	Use stronger smoothing and a much weaker bgAccel_ISF_weight — MA doesn't need early aggressive action.
CGM doesn't allow "SMB always" at cob = 0	Dexcom and Libre 3 both work. MA has cob > 0 immediately after bolus, avoiding this edge case.
Very low hourly basal	No need for boosted SMBs — the bolus covers the meal.
Erratic sweet drinks and snacks	Much less of an issue when a bolus is given with the drink; bgAccel_ISF stays dialled softer.

7.3 Optimize your HCL first

Before setting up MA, your HCL needs to be in good shape. The same rules apply as for FCL:

1. Switch off dynamicISF. Ignore Autotune. Make sure your profile parameters are correct.
2. Meal management tuning: check that your ISFs handle the rising bg after your meal bolus loses power.
3. With correct ISFs, expand SMB sizes to 120 minutes of basal.
4. Handle fatty-meal insulin resistance. Two options:
    - An Automation that increases %profile at signs of post-meal fatty acid resistance (see the AAPS readthedocs "Stagnation at high bg values" section).
    - Switch to the autoISF dev variant of AAPS, with only the dura_ISF component enabled (leave the other ISF weights at 0 or off).

Resources for HCL tuning: https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings

Once your HCL is satisfying, proceed.

7.4 Setting up MA with autoISF

Follow Chapters 5 and 6, but with softer settings throughout. You still get a bolus, so autoISF doesn't need to be as aggressive.

Critical differences from FCL settings:

Setting	*FCL recommendation*	*MA recommendation*
`smb_max_range_extension` (Ch. 5)	2.0–3.0	Lower — your bolus covers most insulin need
`autoISF_max` (Ch. 5)	2.0	Lower — say 1.5
`bgAccel_ISF_weight` (Ch. 6)	Tune aggressively	Much softer — aggressive action not needed
`smb_delivery_ratio` (Ch. 5)	0.6–0.7	Keep at 0.5 (especially with remote bolusing)
`iob_threshold_percent` (Ch. 5)	~60% or as tuned	Same discipline — install it, do not skip

> ⚠️ For remote bolusing setups (parent + child via NSClient): do not raise `smb_delivery_ratio` above 0.5. If a parent initiates a bolus remotely while autoISF is also delivering SMBs, an elevated delivery ratio amplifies the risk of insulin overlap.

> ⚠️ Always install your iobTH (Chapter 3, §3.4). MA is safer than FCL in most respects, but that's precisely because it uses fewer aggressive tools — iobTH is still essential.

7.5 Pre-bolussing strategies

Once the basics are in place, the main design decision in MA is: how much to pre-bolus, and how long before the meal.

THE FULL MEAL BOLUS

Same as classic HCL, no carb entry: you give a calculated meal bolus 10–20 minutes before eating. autoISF runs in the background and supplements with SMBs if glucose behaves unexpectedly.

Pros: familiar; minimal thinking; reliable outcomes. Cons: you're still counting carbs mentally, even if you're not entering them. You're doing most of the work that a full FCL would do for you.

When to use it: meals where you're confident of your carb estimate, especially meals with fast absorption.

THE SMALL PRE-BOLUS

A strategy halfway between FCL and full HCL: give a small, fixed pre-bolus (e.g., 2–3U, or 25–50% of your usual meal bolus) 5–10 minutes before the meal. autoISF handles the rest via SMBs.

Pros: less mental effort than counting carbs. Still kicks autoISF into the right zone. Cons: tuning the fixed pre-bolus size is personal. Over-pre-bolusing on a smaller meal can hypo you; under-pre-bolusing leaves autoISF with too much catch-up work.

When to use it: for consistent meal patterns where you can calibrate the pre-bolus once and forget.

CONCLUSIONS ON PRE-BOLUSSING

📖

TO PRE-BOLUS OR NOT TO PRE-BOLUS?

A short developer paper addresses this question, available in the autoISF repo: To prebolus or not to prebolus... (https://github.com/ga-zelle/autoISF).

The tradeoff: a pre-bolus gives you a head start on the glucose rise, but the insulin on board from the pre-bolus creates a "gap" later (zero-temping) that makes late-meal management trickier. What you gain in peak height, you partly lose in hand-over quality from bolus-dominant to SMB-dominant management.

Further investigations by MA users are welcome. The "best" answer depends on your meal variability and pre-bolus consistency.

7.6 Disturbances, sensitivity, and exercise in MA

All the tools from Chapters 7 and 8 apply:

- Manual nudging (§5.3) — odd TT, %profile, bg target.
- Automations (§5.2) — for recurring situations.
- Custom cockpit buttons (§5.4) — for your "DIY cockpit."
- Exercise (Chapter 6) — simpler in MA, because you can just reduce the meal bolus.

For exercise specifically in MA: you handle exercise the traditional HCL way — reduce the meal bolus ahead of exercise. The "exercise after meal" problem from §6.5 is much simpler, because the reduced bolus itself provides the needed iob cap.

You still benefit from dynamic iobTH and the exercise mode for exercise that happens independently of meals.

7.7 MA for children and remote-control setups

Many of the challenges of FCL for children apply to MA as well, but to a lesser degree:

- Sharp sensitivity or circadian changes — set appropriate temporary or scheduled profile percentages; MA is less demanding on profile accuracy than FCL.
- 24/7 Bluetooth discipline — giving meal boli (plus reliable profile basal) reduces the stakes of brief connectivity loss. Install an alarm on the caregiver's phone for disconnections.
- Caregiver coordination — keep `smb_delivery_ratio` at 0.5 as noted above. Communicate about who is bolusing and when.

For kids, MA is often the realistic destination — not a stepping stone to FCL — because the prerequisites for FCL (site discipline, BT discipline, 24/7 vigilance) are genuinely harder.

7.8 Final thoughts

💬

WHERE THE AUTHOR STANDS

I'm genuinely uncertain whether the effort of setting up MA is worth it compared to either (a) a well-tuned vanilla HCL with sloppy carb inputs, or (b) going all the way to full FCL.

MA probably offers a higher safety level than FCL, especially when the full set of FCL prerequisites aren't permanently guaranteed. The reduced daily vigilance may itself be worth it for some users.

Looking at your own situation honestly — what's missing for full FCL, and how long will it take to fix — is the best guide.

Other algorithmic routes that also use Meal Announcement-style workflows are covered in the next chapter (Boost, AIMI, EatingNow, Tsunami).

CHAPTER 8

Other Avenues to Full Closed Loop

This chapter surveys alternatives to autoISF FCL. Each section is intentionally brief — if one of them matches your situation, follow its links for the full story.

8.1 FCL with AAPS Master and Automations (the simple FCL)

Since autumn 2023, AAPS Master has supported Full Closed Loop as an official option. It uses personalized Automations to elevate iob when a meal-related rise is detected — no autoISF required.

Prerequisites are similar to autoISF FCL (see Chapter 1): fast insulin, excellent CGM, stable technical setup, well-tuned HCL as a starting point. Full details: https://androidaps.readthedocs.io/en/latest/Usage/FullClosedLoop.html

What you give up vs. autoISF FCL:

- Less sophisticated meal detection (relies on delta thresholds instead of parabola-fitted acceleration).
- No automatic dura_ISF-style plateau handling (you'd build that manually via Automations).

- Less peak-shaping precision.

What you gain:

- Dramatically simpler setup. No 18 tuning parameters.
- Fewer ways things can go wrong.
- Widely published results — a randomized cross-over study (PubMed: *First Use of Open-Source Automated Insulin Delivery AndroidAPS in Full Closed-Loop Scenario; Pancreas4ALL Randomized Pilot Study*) and an ongoing 2025 Australia/NZ study (CLOSE IT trial protocol) both show respectable outcomes.

A useful hybrid: run the autoISF dev variant of AAPS with `"Enable ISF adaptation by glucose behavior"` OFF. You keep the benefits of `smb_max_range_extension` and `smb_delivery_ratio > 0.5` for stronger SMB boosting, but drive the whole system via Automations rather than autoISF's ISF modulation. See the comparison in Case Study 13.1.

This method is strongly recommended for users who want an entry into FCL without the full autoISF set-up project. Many users achieve ~80% time-in-range this way.

8.2 FCL with dynamicISF

dynamicISF is available in AAPS (dev variants) and in Trio/iAPS. Unlike autoISF's bgAccel_ISF, it does not boost SMBs on detected acceleration — it's a slower, bg-level-driven ISF modulator, conceptually closer to Autosens but with much more dynamic range.

For FCL use: dynamicISF can be stretched into an FCL tool, but it has an inherent timing disadvantage — it responds to bg level, not to acceleration or delta. That means it's late to the party for meal detection.

It may suit users who:

- Have strong sensitivity swings they can't proactively predict.
- Don't want to tune 18 autoISF parameters.
- Are comfortable with an FCL that gets to the right answer a bit late.

It does not suit users who:

- Want best-in-class peak height control.

- Have a high-carb, fast-absorbing diet.

Case Study 13.2 would be a valuable addition — if you use dynamicISF for FCL successfully, please submit one with a week of 24h scatter data plus an analyzed meal showing when and how dynamicISF built iob.

Resources: - AAPS: https://discord.gg/DfvK5HnxXu (search "dynamicISF") - Trio/iAPS: https://discord.gg/gGKXW5uX3m (section `dynamic-isf-cr`)

8.3 Meal-Announcement dev variants: Boost, AIMI, EatingNow, Tsunami

Several AAPS-derived dev branches implement their own meal-handling approaches. All of these are actively evolving; consult their docs for current state.

BOOST

Boost adds code around SMB calculation with a daily "Boost window" during which the extra logic is active.

- Uses a variant of dynamicISF with predicted-bg weighting (40–75%) to mimic higher sensitivity at lower bg.
- Can trigger early boluses from delta and average-delta accelerations when iob is below a user-set threshold — so it can work with or without meal announcement.
- Max "Boost Bolus Cap" set as a percentage of TDD (default 2.5%, up to 5%) for safety.
- The AAPS default 50% `smb_delivery_ratio` can be overwritten with a higher value ("Boost insulin required percent"), suggested > 75%.
- Automatically stops when delta and average deltas align (steep rise has become a steady rise).

Boost's acceleration-based trigger is conceptually similar to autoISF's bgAccel_ISF route. With an excellent CGM, autoISF detects acceleration slightly earlier and can boost more strongly. Boost is simpler to configure for users who don't want to tune 18 parameters.

See Case Study 13.3 for a Boost-based FCL for a child.

AIMI

AIMI is another dev variant available for AAPS and ported into Trio. Uses a meal-announcement workflow with its own acceleration-based aggressive response.

- https://discord.gg/tPDQzS3Bq3
- Trio port: https://github.com/mountrcg/Trio#aimi-b30

EATINGNOW

EatingNow is another AAPS dev variant with a Meal Announcement workflow. Active development; check the AAPS Discord and associated repos for current status.

TSUNAMI

Tsunami is another AAPS dev variant. Also active development; same guidance — check current community resources.

> 💡 *On all these variants: they're real, legitimate, sometimes excellent — but they're each somebody's active side project. Documentation quality varies. Community support is thinner than for AAPS Master or the autoISF tree. Factor this into your decision.*

8.4 No-bolus looping with precise carb inputs

A philosophically opposite approach: give precise carb inputs (with absorption times) to the loop, but no boluses at all. The loop handles everything via SMBs based on the carb information.

This is the traditional iOS Loop approach, and some "pro" iOS Loop users achieve impressive time-in-range (and time-in-tight-range) performance this way — at the cost of meticulous carb accounting.

Not recommended for casual use. It demands constant mental load for carb estimation and absorption-time input. It does not eliminate meal-time thinking — it just shifts the work from "what size bolus?" to "how many grams, absorbing how fast?"

8.5 Future directions: ML and dual hormones

MACHINE LEARNING / AI

There's ongoing interest in building a self-learning loop. Industry will likely produce a first-generation commercial solution aimed at users with poor HbA1c and weak meal handling — a safe gradual improvement over standard therapy, but not top-tier performance.

A top-performing self-learning system may be genuinely hard to design, because:

- Self-learning relies on past data, which breaks down when you do something different today (fasting day after a feast day, new job schedule, travel).
- Self-learning systems don't teach the user how they work, so the user can't intelligently adjust for situations the system hasn't seen.

This is the opposite of the FCL philosophy in this book — here, the user understands the system and adjusts it deliberately.

DUAL HORMONE SYSTEMS

Many see dual-hormone (insulin + glucagon) FCL as the ultimate system. The glucagon component would cover the largest current limitation: that zero-temping is a weak tool against impending hypos.

What dual hormone could enable:

- More aggressive insulin treatment for highs, because glucagon backs up hypo prevention.
- No weight-gain concerns from roller-coasters (glucagon stimulates glucose release, not caloric intake).
- Potentially dramatically simpler exercise management.

What's in the way:

- Extra device complexity (a second cartridge / pump).
- Significant extra cost.
- Unknown real-world handling by typical users.

📖 *Off-topic note: there's also research on adding very small amounts of a glucagon analogue to an insulin cartridge as an additive to make Lyumjev act even faster. That would be valuable for FCL even before dual-cartridge systems become mainstream. Background: "Insulins, DIA..." PDF in https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings*

8.6 Co-medications that facilitate FCL

GLP-1 RECEPTOR AGONISTS

The drugs most of the world knows as weight-loss medication are also interesting for T1D management, because slowed gastric emptying makes life much easier for any closed loop.

Slower carb absorption:

- Reduces peak heights.
- Tolerates slower insulins.
- Reduces tuning sensitivity — a less aggressive FCL still performs well.
- Opens FCL as a viable option for a broader T1D population.

EMERGING EVIDENCE IN T1D

A small retrospective study of 26 adults with T1D on tirzepatide over 8 months:

- HbA1c reduced by 0.59% at 8 months.
- Body weight reduced by ~10% at 8 months.
- Time-in-range (70–180 mg/dL) increased by ~12.6%.
- Time-in-tight-range (70–140 mg/dL) increased by ~10.7%.
- Time above 180 mg/dL decreased by ~12.6%.
- Well tolerated; only 2 of 26 discontinued.

Source: *Efficacy and Safety of Tirzepatide in Adults With Type 1 Diabetes: A Proof of Concept Observational Study* (J Diabetes Sci Technol 2025;19(2):292–296).

ACCESS

In most markets, T1D patients are not currently eligible for GLP-1 prescriptions — they're approved for T2D and weight management. Access depends heavily on your doctor and region.

Follow the discussion:

- Discord: https://discord.gg/YvrHYmdamc
- Diabettech: https://www.diabettech.com/glp1-ra/glp1-ras-and-type-1-diabetes-a-retrospective-study-of-incretin-based-therapy-use-with-open-source-aid/

If you have experience using a GLP-1 drug with an autoISF FCL, please submit a case study.

8.7 Final remark on alternatives

CHOOSING YOUR PATH

This is an exciting time in the open-source T1D community. You have many viable routes to good looping:

- *Vanilla HCL with sloppy carb inputs — good for many users, needs only modest tuning.*
- *AAPS Master FCL with Automations — simple, well-studied, solid time-in-range.*
- *MA with autoISF (Chapter 7) — middle ground for users missing one or two FCL prerequisites.*
- *Full autoISF FCL (Chapters 5–8) — highest performance ceiling, biggest setup investment.*
- *Boost, AIMI, EatingNow, Tsunami — alternative dev branches with their own design tradeoffs.*
- *Dual hormone systems (future) — potentially transformative, not yet ready.*
- *GLP-1 co-medication — promising, depends on access.*

Pick what fits your life, your temperament, and your access constraints. You don't need to justify your choice to anyone — just have a safe, interesting, and overall enjoyable journey.

And just eat.

PART VI

Monitoring & Troubleshooting

⁂

Keeping the loop healthy once it's running, and fixing it when something goes wrong.

CHAPTER 9

Performance Monitoring

If you followed the setup chapters, you've earned something important: the right to let the loop run. The whole point of FCL is to get there and then stop thinking about it.

This short chapter is about how to keep an eye on the system without slipping back into daily over-tweaking.

9.1 Everyday life with your FCL

The bar to clear is not perfection. The bar is *"mostly good, with occasional needs for a small intervention."*

A well-tuned FCL will:

- Handle most meals hands-off, with glucose staying in range (70–180 mg/dL).
- Occasionally run a bit high or a bit low — small deviations are normal.
- Require a small snack every so often when the insulin tail threatens to bring you low.
- Ask for manual help only for truly out-of-spectrum events (unusual snacks, sick days, heavy exercise).

Resist the urge to fix what isn't broken. If your weekly time-in-range is holding, leave the settings alone.

PRE-MEAL RHYTHM MATTERS

What your bg and iob look like *before* a meal predicts how well the meal will go.

State at meal start	*Outcome*
Glucose rising; iob negative or near zero	Bad. Expect a high peak.
Glucose flat at target; iob near zero	OK.
Glucose slightly declining; mildly positive iob	Good. Loop is "loaded" for the meal.

An EatingSoonTT (§3.5) or the automated low-TT pattern can help produce the third state consistently. Also:

- A small appetizer (olives, soup, a handful of nuts) — pushes bg slightly up in a predictable way the loop can respond to early, kicking off SMBs before the main course.
- A late small dessert — dark chocolate, a cracker with cheese — uses the insulin tail that's "lurking" from the main meal. Better than letting the tail hypo you.

Over time you'll develop intuition for your own rhythm. A little mindfulness here pays off more than fine-tuning any parameter.

9.2 What to watch — and when

Rolling weekly data, not daily. Daily data is too noisy to act on.

THE CORE THREE VIEWS

1. xDrip 7-day 24-hour scatter diagram. The single fastest way to spot patterns. If you see a consistent cluster of outliers at a specific time of day, you have something to investigate.
2. Nightscout Reporter (or equivalent) — weekly time-in-range, hypo counts, daily curves. Useful for tracking trend direction.
3. AAPS home screen — real-time situational awareness during the day. Don't stare at it, but glance.

WHEN %TIR STARTS DRIFTING

If your weekly TIR drops, ask in this order:

1. When are the outliers happening? Time of day, relation to meals, relation to sleep.
2. Is a specific meal or snack habit to blame? Are there "bad" meals and "good" meals?
3. Is there a pattern with exercise or hormonal/disease cycles? Period week? A strength training day you added?
4. Does it correlate with infusion site age, or CGM day 1? "Bad" cannulas and new sensors can masquerade as a tuning problem.
5. Have you changed anything recently? A new Automation, a profile tweak, a diet change? Start by undoing it.

Many "outlier days" trace back to occlusions, leaky cannulas, or CGM inaccuracy — nothing to do with your autoISF settings.

Before changing any settings, rule out the prerequisites (Chapter 10, §10.2).

9.3 When to tune, when not to

DO TUNE WHEN…

- A lifestyle change is persistent: new job, new sleep schedule, pregnancy, sustained weight change.
- A circadian pattern has genuinely shifted (validated over a couple of weeks, not one noisy week).
- You've identified a specific profile ISF hour that's clearly wrong via open-loop testing or emulator analysis.

DON'T TUNE WHEN…

- You had one bad meal last Tuesday.
- You see a single outlier on the xDrip scatter plot.
- You're curious whether an 8% weight change would improve things.

> ⚠️ Case Study 8.2 (Futility of tuning based on one extreme meal) — a negative example where tuning to optimize one unusual meal destroyed performance on normal meals. Read it if you're tempted.

> 💡 *Settings do not have to be great for any one event. They must suit "good enough" most of what you encounter. Chasing perfection for rare events will cost you time-in-range on the meals you eat every day.*

IF YOU MUST TUNE

Change one thing. Wait a week. Look at the full spectrum of your meals, not just the one you were worried about. Only then decide whether it helped overall.

The Emulator (Chapter 11) is invaluable for this — it lets you ask "what would last week have looked like with this setting changed?" without actually changing anything.

9.4 A sane review cadence

- Weekly: glance at xDrip 7-day scatter. If nothing jumps out, you're done.
- Monthly: look at weekly %TIR trend. Is it stable, drifting, or improving?
- Quarterly: review your Automation list. Shelve what you aren't using. Consider whether any recurring manual intervention should become an Automation.
- Annually or on major life changes: re-check that your profile ISFs still match reality via open-loop testing.

That's it. More frequent tuning almost always hurts.

9.5 Enjoy

> 💬 *If you can develop a mindfulness while remaining relaxed and positive-looking ahead, that may be the best recipe for good success — not every time, but more and more often.*

Chapter 10 is for when something does go wrong.

CHAPTER 10

Troubleshooting

When the loop stops behaving, work through this chapter in order. Most problems are not tuning problems — they're prerequisite problems, and you'll find them quickly if you check systematically.

10.1 First: know how to get back into HCL

Your escape route is your most important safety tool. You should be able to execute it from muscle memory.

TURNING FCL OFF

Two paths:

1. Preferences → OpenAPS SMB → autoISF → "Enable ISF adaptation by glucose behavior" = OFF. Fastest shutdown.
2. Tap the loop icon (violet FCL → green HCL) if your AAPS version supports this UI. Automatically re-enables the overview buttons.

AFTER TURNING IT OFF

- Re-enable the insulin button: Preferences → Overview → Buttons → Insulin = ON.
- Start bolusing for meals yourself again.

- Note that HCL-after-FCL runs standard oref SMB+UAM without autoISF — even if your autoISF weights are still tuned. That is the safe default. Re-enabling autoISF for HCL is a separate project.

PARTIAL FALLBACK

You don't have to go all or nothing. A common arrangement:

- FCL for dinners (when you have time to watch it).
- HCL for breakfast and lunch (where you're rushed anyway).

Use a time-window Automation to control "Enable ISF adaptation by glucose behavior." The loop icon will show the current state (green for HCL, violet for FCL).

10.2 Check the prerequisites

Most FCL problems are actually prerequisite problems. Start here.

TECHNICAL PREREQUISITES

☐ Bluetooth stable? Pump connection alerts firing? Phone always within range?

☐ CGM quality? Was this sensor fine yesterday and is jumping today? Is it day 1 of a new sensor (often noisier)?

☐ Cannula / pod age? Is it past your reliable lifetime (often 48h, not the manufacturer stated)?

☐ Were alerts / alarms silenced recently?

DATA-QUALITY CHECKS

Useful: analyze data excluding:

- Sensor day 1.
- Cannula/pod days > 2.

If excluding those "bad" days makes your data look fine, the problem isn't autoISF — it's a prerequisite.

DID YOU FOLLOW THE SETUP SEQUENCE?

It's easy to drift over time. Verify:

- ☐ Your SMB range extension is still wide enough (Chapter 3, §3.1).
- ☐ Your iobTH is set, and the odd/even target logic is enabled (§3.4).
- ☐ Your `bgAccel_ISF_weight` and `pp_ISF_weight` values are what you expect (not accidentally shifted by an Automation with no restore tandem).
- ☐ Autosens is OFF, dynamicISF is OFF (if applicable).

PROFILE STILL CORRECT?

A common failure mode: you built your autoISF on top of a profile that has drifted. Questions:

- When did you last validate your ISFs via open-loop testing?
- Has your body weight, lifestyle, or hormonal pattern changed?
- Were you relying on dynamicISF or Autotune to "cover" the profile before FCL exposed the gap?

If the profile is off, no amount of autoISF tuning will fix it. Go back to HCL, test and fix the profile, then return to FCL.

IS IOBTH SET CORRECTLY?

iobTH is modulated dynamically (§3.4). In the AUTO ISF tab, check the "effective iobTH" line. Things that silently change it:

- An active TT.
- An active %profile.
- Exercise button yellow.
- Activity Monitor adjustment.
- Recently-run Automation that set a different `iob_threshold_percent` without a restore tandem.

If your effective iobTH doesn't match your expectation, that's the first thing to fix.

10.3 Glucose goes too high

MEALS RECOGNIZED TOO SLOWLY

Likely causes:

- ☐ Bluetooth dropouts at meal start — phone was across the room when you sat down.
- ☐ CGM quality — jumpy values confused acceleration detection.
- ☐ Try an aperitif — a small appetizer a few minutes before the meal gives the loop time to see an acceleration.

FIRST SMBS ARE DELAYED

- ☐ Blocked by the 30% jump rule? (Chapter 2, §2.4.) Check the AUTO ISF tab for "SMB disabled" messages.
- ☐ Odd-numbered TT or profile target active? An Automation might have set one.
- ☐ Pump connection issue — BT or physical.
- ☐ Phone proximity — was the phone near the pump at meal start?

SMBS ARE TOO SMALL

In rough order of likelihood:

- ☐ Capped by `smb_max_range_extension` — widen it (Chapter 3, §3.1).
- ☐ Capped by `autoISF_max` — widen it (§3.2).
- ☐ `bgAccel_ISF_weight` set too low — raise it cautiously.
- ☐ `pp_ISF_weight` set too low — raise it cautiously.
- ☐ CGM smoothing eating the early acceleration signal — try lighter smoothing.
- ☐ Sensitivity modulation active — Activity Monitor, exercise mode, %profile reducing the insulin you'd otherwise get.
- ☐ ISF adaptation accidentally off after a Preferences tweak or Automation.

See also: https://github.com/ga-zelle/autoISF → "How to get larger SMBs" for a dedicated troubleshooting PDF.

STUCK AT HIGH BG

- ☐ dura_ISF not engaging? Check the AUTO ISF tab during the plateau — is dura contributing?
- ☐ dura_ISF_weight too low — raise it in small steps (watch for late hypos).
- ☐ Fat-related temporary insulin resistance — consider an Automation that sets elevated %profile at high-FPU meals.
- ☐ Hypothesis: iobTH too low — the loop wanted more insulin but was capped.

Often the right fix is not at the stuck-high plateau itself. If bgAccel/pp_ISF had been sharper, the peak would have been lower and shorter, and dura wouldn't need to work as hard. Look upstream first.

10.4 Glucose goes too low

PHANTOM MEAL TRIGGERS

- ☐ Outside usual meal times? Set an odd profile target for those hours.
- ☐ First SMBs too big — reduce `bgAccel_ISF_weight` (cautiously).
- ☐ Snacks triggering full meal responses — set up a snack cockpit button (§5.4).
- ☐ Compression lows creating false "meal start" signals — odd-numbered night target or see Case Study 5.3.

TOO MUCH INSULIN DELIVERED

- ☐ iobTH too high? Check the effective iobTH in the AUTO ISF tab.
- ☐ iobTH-changing Automation ran and didn't restore? Verify the tandem restore Automation worked (§5.5).
- ☐ SMB range extension too wide for your needs — narrow it.
- ☐ **`autoISF_max`** too high — narrow it.
- ☐ **`smb_delivery_ratio`** too high — reduce across the board.
- ☐ One of the ISF weights too aggressive — usually `dura_ISF`, which has a cumulative effect. Check the AUTO ISF tab to see which component is dominating.
- ☐ Temporarily more insulin-sensitive than usual (e.g., after heavy exercise yesterday)? Use a temp %profile switch rather than changing long-term settings.

LATE HYPOS AFTER MEALS

This is the hardest category. Options in order:

1. Check dura_ISF_weight first. It's the most common cause of late hypos.
2. Reduce `bgAccel_ISF_weight` if early SMBs are too big.
3. Identify the meal types where this happens — often low-fibre, low-fat, low-protein meals where the insulin tail outlasts the carb absorption.
4. Take a small snack proactively (5–10g) when the loop warns of an imminent hypo. The loop's carb recommendation is often inflated — you usually need much less than it suggests, because the loop doesn't know about fat/protein still digesting.
5. Adjust your diet — add more fibre, protein, or fat to balance the carb absorption curve.
6. Lower iobTH — if hypos reliably trace to too much iob in the rise phase.
7. Accept the tradeoff. Sometimes a small snack is genuinely the right answer — "better a snack than a hypo" — especially for weight-conscious users who can reduce intake at the meal itself to compensate.

10.5 Glucose goes too high and too low (roller-coasters)

Roller-coasters point to real problems with the setup. They rarely respond to a single parameter tweak.

STEP BACK AND SIMPLIFY

- Is your lifestyle or diet extreme in some way that's pushing the system past its limits?
- Are your %TIR expectations realistic? The book emphasizes moderate expectations.
- Have you accumulated Automations and workarounds that now interact unpredictably?

VERIFY FOUNDATIONS

- [ ] Was your FCL built on true, experimentally-proven ISFs?
- [ ] Did you follow the tuning sequence — Chapter 3 settings, then `bgAccel_ISF` first in Chapter 4?
- [ ] How often have you looked at the AUTO ISF tab or the Emulator to understand what the loop is actually doing?

WHEN TO START OVER

With many interacting parameters and Automations, you can reach a state where the safest path is not to patch, but to restart:

1. Full restart of autoISF settings — go back to Chapter 3 defaults and rebuild. This is not a failure; the book explicitly expects this for some users.
2. Fall back to HCL — either vanilla AAPS HCL, or HCL with just dura_ISF active for fatty-meal management.
3. Switch methods — try AAPS Master FCL with Automations (Chapter 8, §8.1), which is simpler.
4. Temporarily switch to a commercial system — wait for better tools, then re-evaluate.

None of these is a defeat. They're different valid choices for where you are right now.

10.6 Staying out of trouble

A few principles that prevent most troubleshooting from ever being needed:

1. Interference is the enemy. Let the loop loop. Every user bolus and every additional carb entry distorts the curve autoISF is reading.
2. Don't optimize for single events. The goal is "good-enough through nearly all scenarios." (See Case Study 8.2.)
3. Keep it simple. Resist the latest trick until you understand how it affects your existing balance.
4. Use FCL selectively. It's fine to run FCL only for some meals, or only on certain days. Don't force the loop into situations where you already know it will struggle.
5. Stay connected to the community. The Discord channels are where problems get diagnosed fastest: https://discord.gg/tamvhh57Xs
6. Keep perspective on %TIR expectations. Many non-looping T1Ds are happy above 60% TIR. A study of 16 AAPS users achieved ~80% TIR on a simpler FCL with no meal announcement. These are reasonable reference points. You don't have to chase 95%.

10.7 Just eat

When all else is working — or working well enough — the best thing you can do is stop worrying about it. That's what FCL is for.

* * *

💡 *A note on scope. The Emulator is optional. You can run a successful FCL without ever installing it. The reason to read these chapters is that the Emulator transforms tuning from "wait two days to see if the change helped" into "watch what last week would have looked like with this change, in 30 seconds." If you like data analysis, it is transformative. If you don't, stick with the AUTO ISF tab and skip this.*